

Multi-Agent Financial Research System

Financial Extraction Agent Design & Logical Architecture

1. Overview of the Financial Extraction Agent

The Financial Extraction Agent (Agent #1) is responsible for extracting structured financial information from uploaded financial documents such as annual reports, quarterly reports, 10-K filings, 10-Q filings, balance sheets, income statements, cash flow statements, investor presentations, earnings reports, invoices, and other financial documents. The agent automatically processes multiple document formats including PDF, DOCX, XLSX, CSV, TXT, and images using intelligent text extraction, OCR, document parsing, and Large Language Models (LLMs).

The extracted information is transformed into a standardized JSON structure containing key financial metrics, company metadata, reporting details, and confidence scores. This structured output becomes the foundation for downstream agents such as the Research Agent, Comparison Agent, and Visualization Agent.

The Financial Extraction Agent combines native document parsing, Optical Character Recognition (OCR), document preprocessing, and AI-powered information extraction to ensure high accuracy while preserving the original financial context.

2. Objectives

The Financial Extraction Agent is designed with the following objectives:

- Automatically extract financial information from multiple document formats.
 - Support both digitally generated and scanned financial documents.
 - Extract structured financial metrics into a standardized JSON format.
 - Identify company information, reporting periods, currencies, and financial statements.
 - Normalize financial values, dates, and numerical formats.
 - Handle incomplete or partially structured financial documents.
 - Generate confidence scores for extracted financial data.
 - Provide reliable structured data for downstream AI agents.
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3. Logical Architecture & Workflow

The Financial Extraction Agent follows a structured workflow for processing every uploaded document.

Step 1: Document Upload

The user uploads one or more financial documents through the application interface.

Step 2: File Validation & Type Detection

The system validates the uploaded file, determines its format, verifies supported document types, and extracts document metadata.

Step 3: Native Text Extraction

For digital documents, the agent extracts text using document-specific parsers such as PDF readers, spreadsheet parsers, Word document parsers, and text processors.

Step 4: OCR Processing

If the uploaded document is scanned or image-based, OCR is automatically applied to extract readable text from images.

Step 5: Document Preprocessing

The extracted text is cleaned, normalized, deduplicated, and organized into structured chunks while preserving financial context.

Step 6: Financial Information Extraction

The processed document is provided to the Language Model, which extracts financial metrics, company information, reporting periods, and supporting metadata according to a predefined schema.

Step 7: Data Validation & Normalization

Extracted values are validated, normalized, merged across document chunks, and assigned confidence scores.

Step 8: Structured JSON Generation

The final structured financial data is returned in JSON format together with extraction metadata and confidence scores for downstream processing.

4. Extraction Strategy

To ensure accurate and reliable financial data extraction, the Financial Extraction Agent follows a strict extraction strategy.

System Instructions

- Extract only information explicitly available in the uploaded document.
- Do not assume or infer missing financial values.
- Preserve original currencies, reporting periods, and numerical scales.
- Normalize financial metrics into predefined standardized fields.
- Return **null** whenever required information is unavailable.
- Maintain consistency across all extracted financial metrics.
- Generate confidence scores for every extracted value.
- Preserve supporting source text wherever applicable.

This approach minimizes hallucinations and ensures consistent, reliable structured outputs.

5. Financial Extraction Pipeline

The Financial Extraction Agent implements the following extraction workflow:

1. User uploads financial document.
 2. File type is detected and validated.
 3. Metadata is extracted.
 4. Native text extraction is performed.
 5. OCR is applied when required.
 6. Extracted text is cleaned and normalized.
 7. Document is divided into manageable chunks.
 8. Language Model extracts structured financial information.
 9. Chunk-level results are merged and validated.
 10. Confidence scores are calculated.
 11. Final standardized JSON output is generated.
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6. Key Features

The proposed Financial Extraction Agent provides several important features:

- Multi-format financial document support
 - Automatic document type detection
 - Native PDF, DOCX, XLSX, CSV, and TXT parsing
 - OCR support for scanned documents and images
 - Intelligent document preprocessing
 - AI-powered financial information extraction
 - Financial metric normalization
 - Confidence score generation
 - Structured JSON output
 - Error handling and validation
 - Batch document processing
 - Scalable architecture for enterprise integration
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7. Advantages

The proposed architecture offers several advantages:

- Automates financial data extraction from diverse document formats.
 - Reduces manual data entry and processing effort.
 - Improves extraction accuracy using AI and OCR.
 - Supports both scanned and digital documents.
 - Produces standardized structured financial data.
 - Enables seamless integration with downstream AI agents.
 - Provides confidence-based validation for extracted information.
 - Offers scalable architecture suitable for enterprise financial systems.
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8. Conclusion

The Financial Extraction Agent provides a scalable and reliable solution for automated financial document processing. By combining intelligent document parsing, OCR, preprocessing, structured AI-based extraction, and confidence-driven validation, the system converts unstructured financial documents into standardized machine-readable data.

This milestone establishes the logical foundation for the financial extraction pipeline and enables subsequent agents within the Multi-Agent Financial Research System to perform research, comparison, analytics, and visualization using accurate and structured financial information.